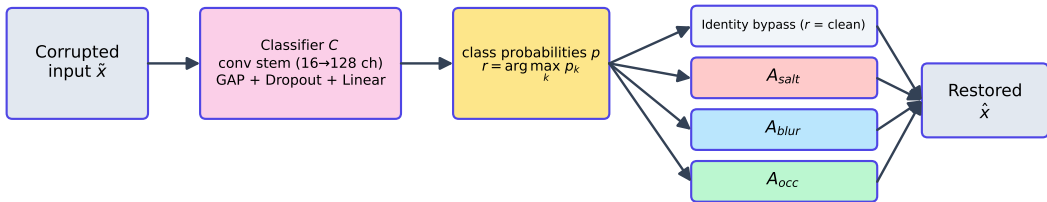
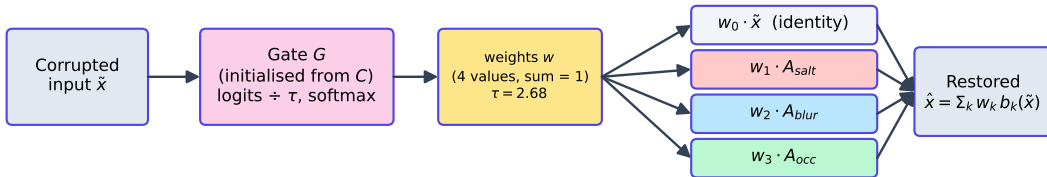


Task 2 - hard routing (classifier + specialists)



Specialists share one architecture (Task 1 autoencoder, 48→384 ch, bottleneck 32) but have independent weights

Task 3 - soft mixture of experts (jointly trained)



Loss: $\lambda_1 \mathcal{L}_{L1} + \lambda_s (1 - \text{SSIM}) + \lambda_c \mathcal{L}_{CE} + \lambda_b \mathcal{L}_{\text{balance}}$; stage 1: experts frozen, gate only; stage 2: all unfrozen, smaller lr